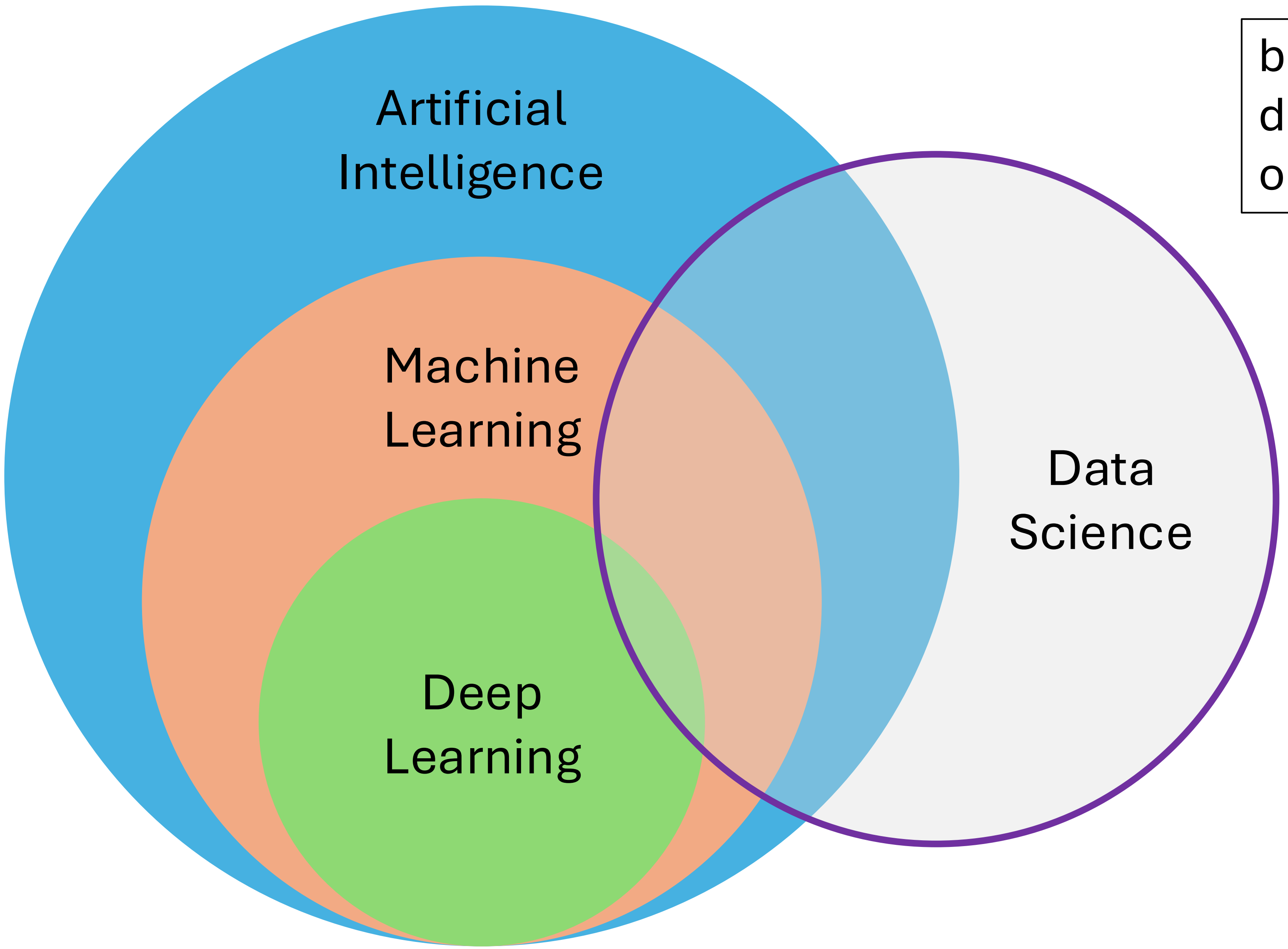
Introduction

Data Science

Course Schedule

1. Exploratory Data Analysis
2. Clustering
3. Probabilistic Models
4. Statistical Inference
5. Machine Learning
6. Feature Engineering
7. Model Evaluation
8. Deep Learning
9. Time Series
10. Causality



blend of diverse components from different domains (statistics, optimization, computer science, ...)

Deep Learning: special kind of ML methods using *deep* neural networks (e.g., CNNs, transformers)

Data Science:
extract knowledge from data (by means of ML, among other things)

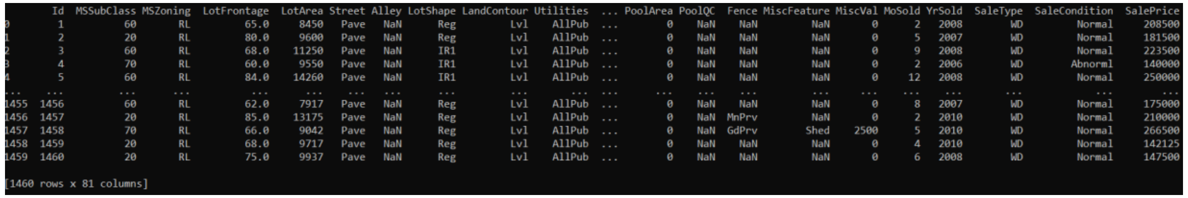
Typical Data Science Workflow

Business Understanding
→ Data Collection
→ Data Exploration
→ Data Cleaning & Preparation
→ Modeling
→ Evaluation
→ Deployment & Monitoring

iterative process

Main Areas of Artificial Intelligence (AI)

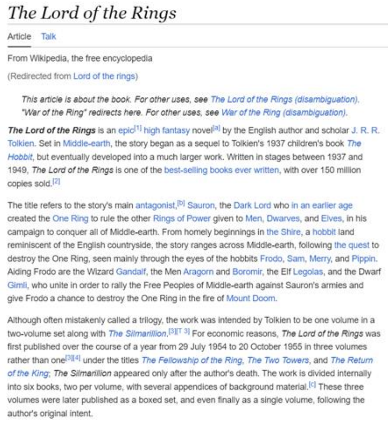
tabular data



	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2008	WD	Normal	208500
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	5	2007	WD	Normal	181500
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	9	2008	WD	Normal	223500
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	2006	WD	Abnorml	140000
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	12	2008	WD	Normal	250000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1455	1456	60	RL	62.0	7917	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	8	2007	WD	Normal	175000
1456	1457	20	RL	85.0	13175	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	MnPrv	NaN	0	2	2010	WD	Normal	210000
1457	1458	70	RL	66.0	9042	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	GdPrv	Shed	2500	5	2010	WD	Normal	266500
1458	1459	20	RL	68.0	9717	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	4	2010	WD	Normal	142125
1459	1460	20	RL	75.0	9937	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	NaN	0	6	2008	WD	Normal	147500

[1460 rows x 81 columns]

language models



The Lord of the Rings

Article Talk

From Wikipedia, the free encyclopedia

(Redirected from Lord of the rings)

This article is about the book. For other uses, see The Lord of the Rings (disambiguation).
"War of the Ring" redirects here. For other uses, see War of the Ring (disambiguation).

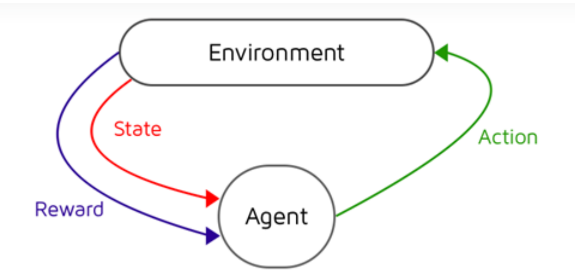
The Lord of the Rings is an epic[1] high fantasy novel[a] by the English author and scholar J. R. R. Tolkien. Set in Middle-earth, the story began as a sequel to Tolkien's 1937 children's book *The Hobbit*, but eventually developed into a much larger work. Written in stages between 1937 and 1949, *The Lord of the Rings* is one of the best-selling books ever written, with over 150 million copies sold.[2]

The title refers to the story's main antagonist,[b] Sauron, the Dark Lord who in an earlier age created the One Ring to rule the other Rings of Power given to Men, Dwarves, and Elves, in his campaign to conquer all of Middle-earth. From homely beginnings in the Shire, a hobbit land reminiscent of the English countryside, the story ranges across Middle-earth, following the quest to destroy the One Ring, seen mainly through the eyes of the hobbits Frodo, Sam, Merry, and Pippin. Aiding Frodo are the Wizard Gandalf, the Men Aragorn and Boromir, the Elf Legolas, and the Dwarf Gimli, who unite in order to rally the Free Peoples of Middle-earth against Sauron's armies and give Frodo a chance to destroy the One Ring in the fire of Mount Doom.

Although often mistakenly called a trilogy, the work was intended by Tolkien to be one volume in a two-volume set along with *The Silmarillion*.[3][T 3] For economic reasons, *The Lord of the Rings* was first published over the course of a year from 29 July 1954 to 20 October 1955 in three volumes rather than one[3][4] under the titles *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*; *The Silmarillion* appeared only after the author's death. The work is divided internally into six books, two per volume, with several appendices of background material.[c] These three volumes were later published as a boxed set, and even finally as a single volume, following the author's original intent.

computer vision

control

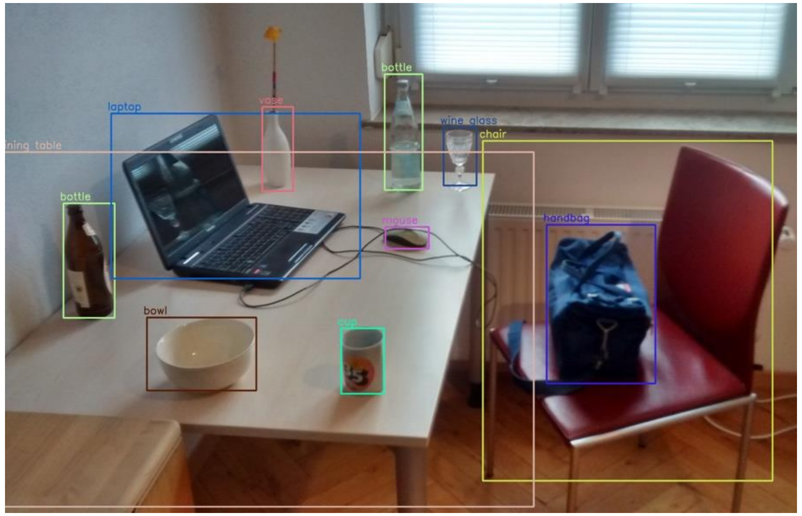


all empowered by one key component:
learning from data → Machine Learning (ML)

When to Use ML (Learning from Data)

automation

too complex for rules



object recognition, chatbot, ...

uncertainty

too complex for humans

AlphaFold

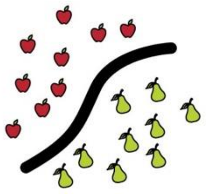
protein structure predictions, demand forecasting, ...

MACHINE LEARNING

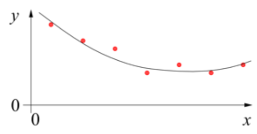
training target available (labeled or past data)

SUPERVISED

CLASSIFICATION



REGRESSION



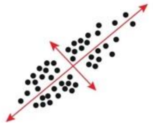
data not labeled in any way

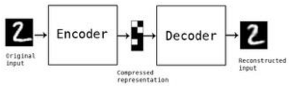
UNSUPERVISED

CLUSTERING

SELF-SUPERVISED

DIMENSIONALITY REDUCTION





ASSOCIATION

no supervision, but goal-based interaction with environment

REINFORCEMENT LEARNING

LEARN STATE OR ACTION VALUES

LEARN POLICY DIRECTLY

learning by teacher (high-dimensional curve fitting)

learning by observation (pattern recognition)

learning by trial-and-error (sequential decision making)

Used Programming Language: Python

- good compromise between rapid prototyping and production
- vast ecosystem
- very popular for data processing and ML: scientific Python stack

Scientific Python Stack

NumPy

pandas

Polars

statsmodels

scikit learn

matplotlib

seaborn

PyTorch

Programming Environments

locally, best use

- a virtual environment to flexibly install packages (e.g., venv)
- an IDE of your choice (e.g., VS Code)
- both plain Python files or Jupyter notebooks are fine

but cloud-based environments also fine (e.g., Google Colab)

Literature

- An Introduction to Statistical Learning
- The Elements of Statistical Learning
- Pattern Recognition and Machine Learning
- Machine Learning

- Deep Learning
- Understanding Deep Learning

Exam form – Assignment (A)

- **Two assignments** during lecture time
- Working in **groups of two** (exceptional three) **students**
- An **assignment** consists of …
  - a specific task given after deadline of registration
  - processing and completing the task
  - presentation (20-30 min) + oral discussion (10-20 min)

Schedule

week	date	topic
4	17.04.2026	deadline of exam registration via QIS
5	20.-24.04.2026	announcement of assignments, groups & presentation dates
8	11.-15.05.2026	no lectures for final preparation of assignment 1
9	18.-22.05.2026	assignment 1 (5 blocks including Friday (22.05.) block 4 and 5)
13	15.-19.06.2026	no lectures for final preparation of assignment 2
14	22.-26.06.2026	assignment 2 (5 blocks including Friday (26.06.) block 4 and 5)